



ENT-164 · Intro to Making



Derby Entrepreneurship
Center at Tufts

CLASS 1 · WEEK 1 · FALL 2026

Introductions and Course Overview

Welcome to Intro to Making. Today we meet each other, see how the semester works, and look at what you'll be able to build by December.

Thu, Sep 10

JCC 301 / Nolop

Hands-on from day one

No experience needed



Thomas Van de Velde · Instructor · Alexander Beattie · TA · Patrice Pinardo · TA

Tufts University · Gordon Institute

What you'll get out of **this class**

01

Hands-on making

Harness technology through hands-on making — no fear, and no prior physical making or coding knowledge required.

02

Maker tools proficiency

Develop confidence in concept drawing, hand modeling, 3D printing, laser cutting, electronics, and connected devices.

03

Team collaboration

Know how to collaborate in multidisciplinary teams to design and build working prototypes.

04

Entrepreneurial mindset

Apply the “build–measure–learn” loop through failing fast and doing rapid iterations.

Let's get to know **each other**

Everyone answers one of these — there are no wrong answers.

01

Why are you taking this class?

02

Tell us about something you made that you're proud of.

03

What would you like to make in this class?

If you see yourself here, **you belong**

Aspiring entrepreneurs

Students looking to launch or work in early-stage tech ventures who want to build proof of concepts to vet ideas with customers.

Future inventors

Those interested in exploring ideas, understanding how things work under the covers, and experimenting with new technologies.

Hands-on team players

Individuals eager to collaborate in cross-functional teams and build real prototypes to test ideas rapidly.

Problem solvers & innovators

Students drawn to building tangible products to prove early concepts and gain traction for their ideas.

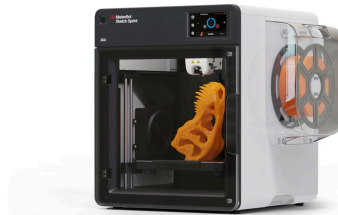
From idea to **working prototype**

Laser cutting, 3D printing, and electronics — the core maker skills you'll use all semester.



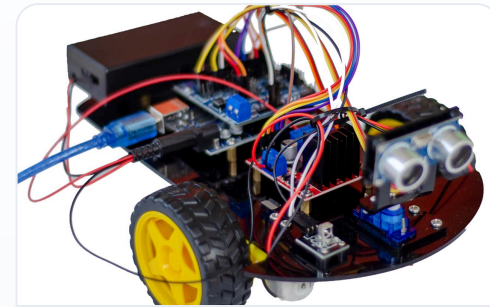
Laser cutting

Turn 2D designs into precise, structural parts.



3D printing

Model and print custom mounts, brackets, and parts.



Robots

Wire motors and sensors, then program the behavior.

Built by students, **one week at a time**

Every project is an excuse to build, test, and learn as a team.



Build and test

Working prototypes, tested in the classroom every week.



Product evolution

Iterate from cardboard mockups to a finished prototype.

Your **instructor** team



INSTRUCTOR

Thomas Van de Velde

Lecturer in Entrepreneurship,
Gordon Institute at Tufts University
· Founder, TeachPilot · Advisor,
Greenway Institute · ex-Amazon,
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TEACHING ASSISTANT

Alexander Beattie

Major: Mechanical Engineering
Class: 2027



TEACHING ASSISTANT

Patrice Pinardo

Major: Computer Science
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Alexander Beattie · Teaching Assistant

Mechanical Engineering · Class of 2027



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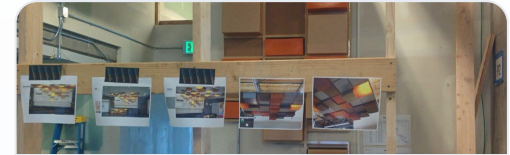
A maker's story: from sketch to scale



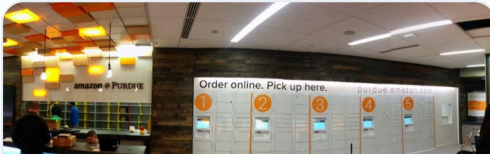
The idea starts as a sketch



The first prototype in use



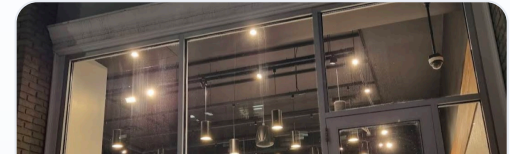
Building the real thing



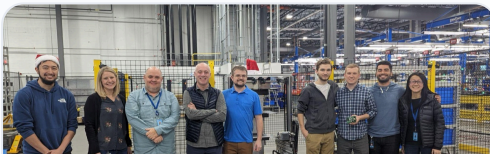
First launch: Amazon @ Purdue



Scaling to thousands of lockers



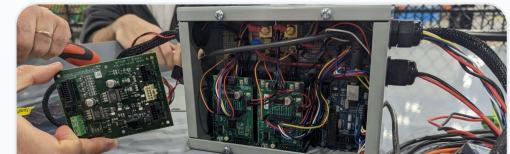
From prototype to storefront



The teams behind the build



Robotics at warehouse scale



Boards, sensors, and wiring

How a typical class **works**

PART 1

60–90 minutes

In the classroom

- Last class recap
- Foundational maker skills
- Short exercises

BREAK

10–15 minutes

Walk to Nolo

PART 2

60–90 minutes

In the makerspace

- Apply maker skills
- Team project work

ASSIGNMENTS

Individual — make small things · **Team** — Robot Challenge (first half) and Final Project (second half)

Office hours & getting help

Stuck on something? Ask early — that's what we're here for.



BOOKING PAGE

go.tufts.edu/meet_tvdy

Book time with the instructor for help with projects, assignments, or anything else on your mind.



CONTACT

thomas.van_de_velde@tufts.edu

Email works too. Teaching assistants are available by appointment.

Attendance & participation

EXPECTATIONS

- ? Actively participate in class activities and teamwork
- ? Email me **before** class if you have a valid reason for missing
- ? Inform your teammates if you can't make a team meeting
- ? Team participation is evaluated via peer reviews

KEY MESSAGES

- ? This class simulates a **real-world work environment**
- ? Don't hesitate to ask for help if you fall behind
- ? Teamwork can be challenging — you're here to learn how to work effectively as a team

What we'll cover, **step by step**

FOUNDATION

Physical making

Nolop tools

CAD design

Laser cutting

3D printing

Microcontrollers

Electronics

MicroPython

Motors

Sensors

PROJECT-SPECIFIC TOPICS

AI integration

Wireless communication

Cloud services

Visualizations

App integration

And more...

Ideate

Weeks 1–4

**Robot Challenge**

mid-semester

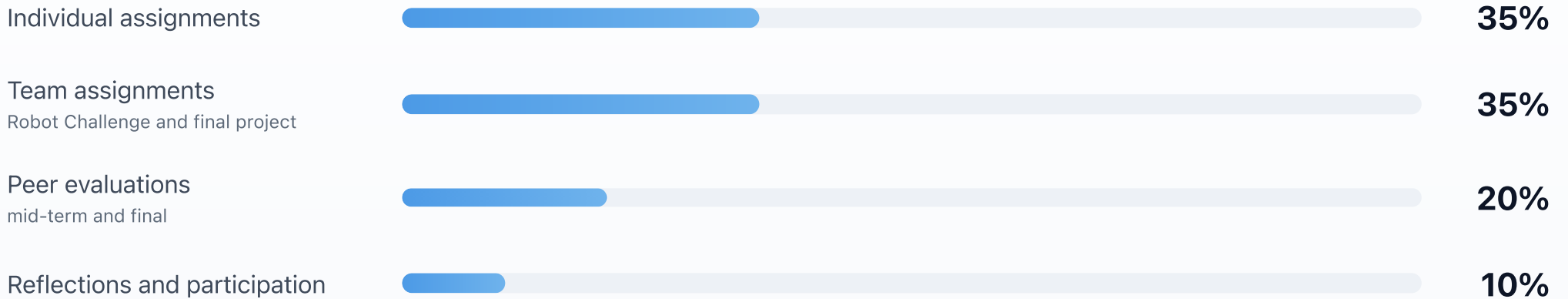
**Test, iterate, refine**

Weeks 9–12

**Demo!**

end of semester

How your grade **adds up**



Team grades are shared by default, but may be adjusted based on individual performance.

The engine of learning in **project-based work**

WHAT IS IT?

A structured pause to examine your work, decisions, and learning process — turning experience into insight.

WHY IT MATTERS

Without reflection, we repeat mistakes and miss growth opportunities.

Experience

1

Do the work

Conceptualize

3

Extract insights

Reflect

2

Analyze what happened

Apply

4

Improve next iteration

KEY REFLECTION QUESTIONS

- ? What worked well and why?
- ? What would I do differently?
- ? What did I learn about myself?

Five times this semester — at the milestone that closes each phase.

What you'll **build with**

And more tools will be introduced throughout the course.



OnShape

Web-based 2D and 3D design tool for CAD modeling.



PrusaSlicer

3D slicer software that prepares your models for printing.



Thonny

Development environment for MicroPython on your devices.



AI tools

Used for software development on devices — a core maker skill.

Everything you need **to start building**

Microcontrollers

The brain of every project you'll build this semester.

Sensors

Light, distance, motion, and more — to sense the world.

Wires & breadboards

Connect components and prototype circuits quickly.

Components

Motors, LEDs, buttons, resistors, and a ton more...

MAKERSPACE

NOLOP

A makerspace open to everyone at Tufts University — laser cutters, 3D printers, electronics benches, and hands-on help.

nolop.org



BEFORE YOU GO

Questions?

Then it's time to make something. Next week: Ideation and Prototyping — we form teams and kick off the Robot Challenge.

WHERE TO FIND US

Office hours

go.tufts.edu/meet_tvdv

Email

thomas.van_de_velde@tufts.edu

Course materials

tuftsmaker.github.io/ENT-164